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CHEMICAL MECHANISMS OF NO_x FORMATION FOR GAS TURBINE CONDITIONS

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Crucial reactions for NO_x and CO were identified, expressions for their kinetics were improved using theoretical predictions, and the improved mechanism was exploited by probing a data set from high-pressure turbulent combustion with a two-reactor model. Reaction-path analysis and linear sensitivity analysis were combined to identify the reactions that were most important in determining NO_x and CO concentrations in lean (ϕ = 0.6), premixed, laminar methane/air flames ranging in pressure from 1 to 40 atm. These reactions were then analyzed using Quantum-RRK techniques and literature data. Revised rate constants and additional reactions were proposed. Analysis revealed that thermal or Zeldovich formation of NO from N_2 + O apparently proceeds via the electronically excited $O(^1D)$ state, not by direct reaction with ground-state $O(^3P)$.

The chemical mechanism was used to model NO_x measurements taken from the burned-gas zone of lean, premixed, turbulent, 5 to 10 atm methane and ethane flames stabilized on a perforated plate. The flow field of the turbulent flames was modeled as a perfectly stirred reactor (PSR) and a plug flow reactor (PFR) in series so that the complex chemistry could be included. Agreement between the model and the data was good, indicating that the PSR/PFR approximation is valid for predicting NO_x from lean, premixed flames. The nitrous oxide (N_2O) pathway was shown to be the dominant route to NO_x formation at lower temperatures (more fuel-lean), with thermal NO_x becoming more important at higher temperatures (closer to stoichiometric). The contribution of prompt NO_x to the total NO_x formed was found to be relatively small under fuel-lean conditions, although it becomes the dominant route to NO_x under fuel-rich conditions.

Introduction

The gas-turbine industry is developing lean, premixed combustors that can burn liquid fuels and natural gas while achieving emissions of less than 25 ppmv NO_x (= NO + NO_2) and CO (dry, 15% O_2 basis). Gas-fired turbines typically operate with preheated air at temperatures up to 620 K (650° F), pressures up to 10 atm, and combustor exit temperatures in excess of 1640 K (2500° F). Aircraft engines operate at even higher pressures and inlet temperatures.

For these reasons, chemical mechanisms of NO_x and CO formation and destruction are of great interest. Mechanisms have been studied extensively, both theoretically and experimentally,[1] but little of this effort has focused on well-defined systems at temperatures and high pressures representative of gas-turbine conditions. An important uncertainty is that pressure dependence of reaction kinetics can be different than in low-pressure flames.

Glarborg et al.[2] and Miller and Bowman[3] developed a detailed mechanism of CH_4 and NO_x chemistry, comparing their detailed mechanism to premixed low-pressure flames and atmospheric-pressure well-stirred reactors. Drake, Blint, and co-workers[4,5] analyzed pressure dependence of selected reactions and computed species profiles in stoichiometric, premixed, laminar flames at pressures up to 20 atm. Along with Malte and Pratt,[6] Miller and Bowman,[3] and Correa,[7] they found an important role for N_2O chemistry in making NO.

This paper focuses on the chemical mechanisms of NO_x and CO formation and destruction in lean, premixed, high-pressure flames, conditions similar to the those found in gas turbines. Using laminar, premixed flames and the Miller and Bowman mechanism,[3] reactions critical to predictions of NO_x and CO were identified. These reactions were reanalyzed using recent literature data and Quantum-RRK analyses, and rate constants, reactions, and

species were added where necessary. The modified mechanism was then used to model data from premixed, turbulent, high-pressure flames. To include the complete chemical mechanism in the model, the turbulent flow field was approximated as a perfectly stirred reactor (PSR) and a plug flow reactor (PFR) in series.

Reaction Analysis

Temperature and concentration profiles were predicted for premixed, adiabatic, laminar flames of methane/air, similarly to the analysis of Drake and Blint.[5] The major differences were that fuel-lean conditions were examined here and that more extensive analyses were made of kinetics for chemically activated reactions. Combustion of a fuel-lean mixture (fuel equivalence ratio $\phi = 0.6$) was studied in freely propagating flames and burner-stabilized flames at 1, 10, and 40 atm.

Flame structures were computed on a DECstation 5000 using the Sandia Chemkin II package[8] including the PREMIX code for premixed, laminar flames. Sandia thermodynamics and transport-properties databases and the chemical mechanism of Miller and Bowman[3] were used initially. Reaction-path analysis[9] and linear sensitivity analysis[8] were used to identify the most important reactions in the mechanism.

Predictions of NO_x and CO concentrations were affected most by the reactions in Table I. Significantly, this list is dominated by association and dissociation reactions, which have kinetics that may depend on the pressure through the concentration of third-body gas ($[M] = P/RT$ for ideal gases); kinetics for abstractions are pressure-independent. Thermal decompositions or addition/stabilizations like $HO_2 \rightleftarrows H + O_2$ have $[M]$-dependent kinetics at low pressures and high temperatures and $[M]$-independent kinetics at high pressures and low temperatures. However, kinetics of addition/decomposition reactions like $H + O_2 \rightleftarrows O + OH$ are $[M]$-independent at low $[M]$ or P and inversely proportional to $[M]$ at high $[M]$ or P. The absence of a detectable adduct and of pressure-dependent kinetics at low pressures disguises many of the association reactions in Table I.

Calculations of kinetics were made with Bimolecular Quantum-RRK[11] and compared to available literature data such as the NIST kinetics database.[12] The primary input parameters for the Quantum-RRK method are Arrhenius association and dissociation rate constants at the high-pressure limit. Kinetics was left unchanged for some reactions, but in other cases, new reactions, thermodynamics, and species were added to the mechanism (Table II). Details of the calculations are presented elsewhere,[12] but selected changes are summarized below.

$N_2 + O \rightleftarrows N + NO$ or N_2O (+M):

Both these reactions proceed via N_2O^*, but analysis reveals that the first reaction must proceed via the electronically excited $O(^1D)$ state, not by direct reaction with the ground state $O(^3P)$.

Attack on N_2 by O is treated as forming NO directly (the thermal or Zeldovich mechanism of NO formation) or forming N_2O, which forms NO when it reacts with O-atoms (the nitrous oxide mechanism). O-atom has been presumed to be in the triplet, ground state, but the reaction $O(^3P) + N_2(^1\Sigma_g^+) \rightleftarrows N_2O(^1\Sigma^+)$ is spin-disallowed, making it slow. The potential energy surface for the addition is well understood,[17] and the O-addition kinetics are described adequately with Lindemann-Hinshelwood falloff.[5,17]

$O(^1D)$ addition must be the source of N + NO (Tables II and III). $O(^3P)$ addition is too slow, but $O(^1D)$ addition with $k_\infty = 2 \times 10^{14}$ $cm^3mol^{-1}s^{-1}$ matches literature data[13] extremely well if $O(^3P) \rightleftarrows O(^1D)$ is assumed to have been equilibrated in the experiments. The dominant product channel of $O(^1D) + N_2$ is $O(^3P) + N_2$, chemical deactivation occurring in parallel with physical, collisional deactivation of the electronically excited state to the ground state.

TABLE I
Reactions important in Miller and Bowman mechanism for NO_x and CO predictions.

Association and dissociation reactions:		Abstraction reactions:
$H + O_2 \rightleftarrows O + OH$	$N + O_2 \rightleftarrows NO + O$	$H_2 + OH \rightleftarrows H_2O + H$
$H + O_2 + M \rightleftarrows HO_2 + M$	$NO_2 + M \rightleftarrows NO + O + M$	$HCO + OH \rightleftarrows H_2O + CO$
$H + HO_2 \rightleftarrows 2OH$	$N_2O + O \rightleftarrows 2NO$	$HNO + OH \rightleftarrows H_2O + NO$
$CH_3O + M \rightleftarrows CH_2O + H + M$	$NO_2 + \rightleftarrows NO + O_2$	
$HCO + M \rightleftarrows H + CO + M$	$NO_2 + OH \rightleftarrows HO_2 + NO$	
$HCO + O_2 \rightleftarrows HO_2 + CO$	$NO_2 + H \rightleftarrows NO + OH$	
$CO + OH \rightleftarrows CO_2 + H$	$NH + NO \rightleftarrows N_2O + H$	
$N_2O + M \rightleftarrows N_2 + O + M$	$HNO + M \rightleftarrows H + NO + M$	
$N + NO \rightleftarrows N_2 + O$	$N + OH \rightleftarrows NO + H$	

TABLE II
Mechanism modifications for M = N_2. Rate constants fitted to $AT^b\exp(-E/RT)$ in cm^3, mol, s, cal, K units over 300 to 2000 K and 1 to 40 atm.

		A	b	E	Comments:
$O + OH + M \rightleftarrows HO_2 + M$	2.1E18	−1.26	790		Added reaction.
$H + HO_2 \rightleftarrows 2OH$	1.66E14	0.00	880		Slight revision; reaction via H_2O_2*.
$CH_3O \rightleftarrows CH_2O + H$, k_∞	2.E25	−2.7	26700		Added k_∞ retained k_o, and
$CH_3O + M \rightleftarrows CH_2O + H + M$, k_o	1.00E14	0.0	25000		applied Lindemann falloff.
$CO + OH(+M) \rightleftarrows CO_2 + H(+M)$ $HOCO \rightleftarrows CO_2 + H$, $CO + OH$	—	—	—		Parameters and forms of Larson, Stewart, and Golden (Ref. 13).
$N + NO \rightleftarrows N_2 + O(^1D)$	4.6E13	−0.006	290		Added reaction.
$N + NO \rightleftarrows N_2 + O$; i.e, $O(^3P)$	1.4E10	−0.34	510		Revised.
$N + NO + M \rightleftarrows N_2O + M$	2.3E16	−1.2	890		Added reaction.
$N_2O \rightleftarrows O + N_2$, k_∞	1.3E11	0.0	59600		Hanson and Salimian (Ref. 14)
$N_2O + M \rightleftarrows O + N_2 + M$, k_o	6.9E23	−2.5	65000		parameters with Lindemann falloff.
$O + N_2 \rightleftarrows N_2 + O(^1D)$	7.5E16	−1.85	46800		Added reaction.
$O(^1D) + N_2 + M \rightleftarrows N_2O + M$	1.1E24	−2.73	1800		Added reaction.
$O(^1D) + M \rightleftarrows O + M$	6.6E12	0.00	0		For physical deactivation.
$NO_2(+M) \rightleftarrows NO + O(+M)$	—	—	—		Tsang and Herron (Ref. 15) with falloff.
$N + O_2 \rightleftarrows NO + O$	2.2E15	−0.8	7900		Revised; via NOO* decomposition.
$N + O_2 \rightleftarrows NO + O$	6.E17	−1.9	12400		Via NOO* → ONO* decomposition.
$N + O_2 + M \rightleftarrows NOO + M$	2.E20	−1.9	1400		Added reaction.
$NO + O(^1D) + M \rightleftarrows NOO + M$	2.0E17	−2.5	45900		Added reaction.
$NO + O + M \rightleftarrows NOO + M$	3.E18	−1.6	36700		Added reaction.
$NO + O \rightleftarrows NO + O(^1D)$	5.5E14	−0.82	47900		Added reaction; via ONO* (N_2O*).
$NO + O \rightleftarrows NO + O(^1D)$	9.E14	−0.74	47200		Added reaction; via NOO*.
$N + O_2 \rightleftarrows NO + O(^1D)$	1.3E15	−0.8	15200		Added reaction.
$N_2O + O \rightleftarrows N_2 + O_2$	6.6E13	0.00	26600		Tsang and Herron (Ref. 15); via ONNO*.
$N_2O + O \rightleftarrows 2NO$	1.02E14	0.00	28000		Tsang and Herron (Ref. 15); via ONNO*.
$HO_2 + NO \rightleftarrows NO_2 + OH$	1.1E14	−0.71	500		Revised.
$HO_2 + NO + M \rightleftarrows HOONO + M$	1.5E24	−3.5	2200		Added reaction.
$OH + NO_2 \rightleftarrows HOONO$	5.8E28	−6.1	3420		1 atm; added reaction.
	9.1E29	−6.1	3780		10 atm.
	2.2E30	−6.1	4220		40 atm.
$NO_2 + H \rightleftarrows NO + OH$	7.2E13	0.00	10		Revised.
$NO_2 + H + M \rightleftarrows HONO + M$	1.4E18	−1.5	900		Added reaction.
$NH + NO \rightleftarrows N_2O + H$	2.2E18	−1.6	1370		N_2O + H from HNNO*; based on Ref. 16.
$NH + NO + M \rightleftarrows HNNO + M$	1.6E23	−2.6	1820		Added reaction.
$N_2O + H \rightleftarrows HNNO$	8.3E26	−4.7	10540		1 atm; added reaction.
	1.1E27	−4.4	11110		10 atm.
	1.4E25	−3.7	10950		40 atm.
$NH + NO \rightleftarrows N_2 + OH$	1.2E17	−1.5	1300		Added reaction.
$N_2O + H \rightleftarrows N_2 + OH$	5.E15	−0.6	20000		Revised; from $N_2O + H \rightarrow$ NNOH*.

TABLE II (*continued*)

	A	b	E	Comments:
$N_2O + H \rightleftarrows N_2 + OH$	9.5E14	−0.33	19700	Revised; from $N_2O + H \rightarrow HNNO^*$.
$N + OH \rightleftarrows NO + H$	2.E14	−0.39	360	Revised; $NOH^* \rightarrow HNO^* \rightarrow NO + H$.
$N + OH \rightleftarrows NO + H$	4.6E13	−0.29	260	Revised; from $NOH^* \rightarrow NO + H$.
$N + OH + M \rightleftarrows HNO + M$	1.3E16	−1.01	560	Added reaction.

TABLE III
Parameters for N + NO kinetics by Bimolecular Quantum-RRK.

Reaction	A_{fwd}	$E_{a,fwd}$	A_{rev}	$E_{a,rev}$
$N + NO \rightleftarrows N_2O$	3.1×10^{13}	0.2	1.6×10^{16}	114.6
$N_2 + O(^3P) \rightleftarrows N_2O$	3.8×10^{9}	23.2	1.3×10^{11}	59.6
$N_2 + O(^1D) \rightleftarrows N_2O$	2×10^{14}	0	1.5×10^{16}	84.4

A-factors are in cm^3, mol, s units; E_{act}'s are in kcal/mol.

Other input parameters were the Lennard-Jones parameters for the adduct (ϵ/κ=232.4 K, σ=3.828 Å) and the mean frequency $\langle\nu\rangle$ = 996 cm^{-1}.

$O(^1D)$ then is predicted to be the crucial reactant in Zeldovich kinetics. If formed from $O(^3P)$, its concentration may scale with equilibrium or superequilibrium levels of $O(^3P)$, and chemical routes that make $O(^1D)$ directly will exacerbate NO formation. Like singlet CH_2, singlet O must be fully included in detailed mechanisms.

$N + O_2 \rightleftarrows NO + O$ or NO_2 (+M):

"NO_2" usually refers to ONO, but other NO_2 structures also occur. We suggest that N can combine with O_2 to make NOO*, which may either decompose to NO + $O(^3P)$ directly or isomerize to cyclic NOO* and then to NO_2*, which also may decompose to NO + $O(^3P)$. Agreement with the data is good.

$HO_2 + NO \rightleftarrows NO_2 + OH$:

Chemically activated HOONO* is formed. The Quantum-RRK prediction is in excellent agreement with data for both directions.

$NO_2 + H \rightleftarrows NO + OH$:

These reactions form HONO*, which may decompose or be stabilized to HONO. From the NO_2 + H direction, OH + NO are the dominant products, while HONO formation is near its low-pressure limit. In contrast, NO + OH predominantly forms HONO, also in the low-pressure limit. Again, agreement is very good.

$N + OH \rightleftarrows NO + H$:

These reactants were assumed to form NOH*, which may decompose to NO + H or isomerize to HNO*, which may be stabilized or decompose to NO + H. Thermal decomposition of HNO is also described by this reaction set. Predictions were within a factor of three of the data.

$NO_2 + O \rightleftarrows NO + O_2$:

$O(^3P)$ may add to the N atom of NO_2, making NO_3*, or it may combine at the O atom to make ONOO*. In the latter case, exothermic decomposition will form NO + O_2 approximately at k_∞ for the radical combination reaction.

Validation:

In a test against data of Taylor[18] for a 1-atm, fuel-rich, near-adiabatic C_2H_4/O_2 flat flame (ϕ = 1.73), prediction of NO was shown to be dramatically improved. NO predicted by the modified mechanism agreed well with the data, while the unmodified mechanism was high by a factor of two. For every other species, the modified mechanism was slightly better but approximately the same. Both mechanisms did well in predicting H_2, H_2O, CO, C_2H_6, and CO_2. In contrast, predictions of CH_4, C_2H_2,

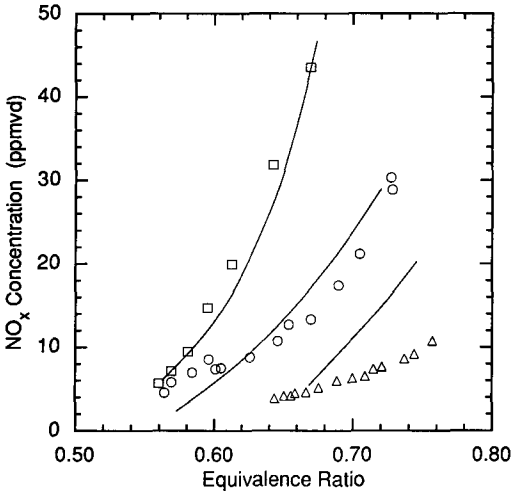

FIG. 1. Comparison of measured and predicted NO_x emissions from premixed methane/air turbulent flames. Solid lines are predictions of the PSR/PFR model. □: P = 10 atm, T_{inlet} = 615 K, residence time = 30 ms, total mass flow = 19.3 g/s; ○: P = 5.1 atm, T_{inlet} = 520 K, residence time = 32 ms, total mass flow = 3.9 g/s; △: P = 5.0 atm, T_{inlet} = 300 K, residence time = 28 ms, total mass flow = 10.3 g/s.

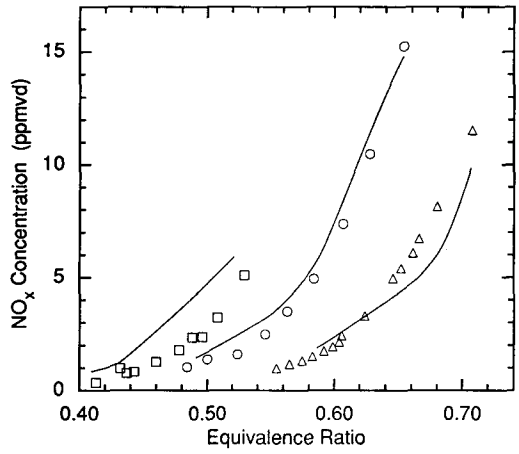

FIG. 2. Comparison of measured and predicted NO_x emissions from premixed ethane/air turbulent flames. Solid lines are predictions of the PSR/PFR model. □: P = 10 atm, T_{inlet} = 610 K, residence time = 27 ms, total mass flow = 10.4 g/s; ○: P = 4.9 atm, T_{inlet} = 520 K, residence time = 37 ms, total mass flow = 5.6 g/s; △: P = 5.0 atm, T_{inlet} = 300 K, residence time = 30 ms, total mass flow = 10.4 g/s.

and C_2H_4 (the fuel) were quite low due to excessive rates of destruction.

Methane flat-flame structures (ϕ = 0.6) were calculated with both mechanisms. Differences from the original mechanism were most significant at higher pressures. At 40 atm, predicted CO was lower by a factor of four while NO was higher by 50% with the modified mechanism.

Experimental

The experimental apparatus and sampling system have been described in detail.[19] Briefly, high-pressure, turbulent, lean premixed flames were stabilized on a 51-mm-diameter perforated quartz disk and confined within a 52-mm-diameter quartz tube extending 150 mm above the quartz disk. This burner was contained in a pressure vessel with quartz windows for visual access. Fuel and air were supplied by cylinders and an inter-cooled, oil-free compressor, respectively. Inlet temperatures as high as 650 K were obtained using a combination of electric heat tapes and a gas-fired preheater. Gases were sampled with a water-cooled stainless-steel probe, dried, and analyzed for NO_x, CO, CO_2, and O_2.

An addition to the system is flame temperature measurement. A Pt-30%Rh/Pt-6%Rh thermocouple (750 μm bead diameter) was inserted down the length of the sample probe, extending 6 mm beyond the probe tip. Measurements were corrected for radiative losses from the thermocouple, typically ~100 K. To avoid catalytic reactions, gas was sampled without the thermocouple in place. The measured rate of temperature drop in the burned gas was 1 to 4 K/ms, depending upon flame conditions. Peak temperatures were up to 100 K less than the adiabatic flame temperature, indicating heat loss to the burner.

Length of the luminous blue-flame zone above the quartz disk was about 25 mm, exceeding 80 mm near blowout. Lean blowout was defined by the ϕ at which measured CO increased dramatically.

The flame was modeled as a perfectly stirred reactor (PSR) and a plug flow reactor (PFR) in series.[19] Physically, the PSR corresponds to the turbulent flame zone, while the PFR corresponds to the post-flame zone; calculationally, the former acts as an ignition source for the latter. A term in the PSR energy equation accounted for heat loss to the burner, and measured temperatures were used in the PFR. PSR size was then chosen to match the experimentally observed lean blowout limit for a given combination of total mass flow, pressure, and inlet temperature. Results were computed on a VAX 6520 using the Chemkin II package of subroutines

and associated programs[5] with the Miller and Bowman mechanism,[3] modified as described above.

Model predictions of NO_x in methane/air flames are compared in Fig. 1 with measurements at inlet temperatures of 300, 520, and 615 K and pressures of 5 and 10 atm. The PSR/PFR model fits the shape and magnitude of the data fairly well. Agreement between the simple PSR/PFR model and ethane/air flame data at the same conditions (Fig. 2) is also good.

Discussion

There must be a recirculation zone to stabilize these turbulent flames, but it will not be perfectly mixed. Given the state-of-the-art in turbulence/chemistry modeling, the choice is either (1) including the details of the flow field and simplifying the chemistry or (2) simplifying the flow-field model and including all of the detailed chemistry. The latter was chosen here. We believe that the volume of the model PSR is related to the volume of the actual turbulent flame zone, which in turn depends on inlet temperature and pressure. Volume of this flame zone decreases as the pressure and temperature increase, both experimentally and in the model.

The model predicts that three major pathways contribute to NO_x formation: thermal (Zeldovich) NO_x, nitrous oxide NO_x, and prompt NO_x[20] (initiated by attacks on N_2 by CH or CH_2). Predictions for the 10-atm methane/air flame with 615 K inlet temperature (Fig. 1) are shown in Fig. 3. At low equivalence ratio (low temperature), the nitrous oxide mechanism is found to be the dominant pathway of NO_x formation. As ϕ and temperature increase, the relative importance of the thermal NO_x pathway increases.

Prompt NO_x is a small contributor to total NO_x under these fuel-lean conditions. However, additional calculations in PSR/PFR geometries show that prompt NO_x becomes the dominant route to NO_x under fuel-rich conditions. Prompt NO_x would be more important in lean combustion of fuels like C_2H_4 and C_2H_2 that yield higher concentrations of CH and CH_2.

The PSR/PFR model predicts that almost all of the NO_x formation occurs in the PSR for the fuel-lean cases. The reason for the decrease in NO_x formation in the PFR zone is that CH, CH_2, and O-atom concentrations drop rapidly once the flow leaves the PSR. At higher temperatures, NO_x formation can continue throughout the PFR, predominantly through the thermal NO_x mechanism. Measurements of the NO_x concentration as function of distance in the perforated-plate combustor support this conclusion. At low equivalence ratio, there was no measurable increase in NO_x with residence time in the burned gas zone. NO_x increased with residence time only at the highest ϕ and temperatures.

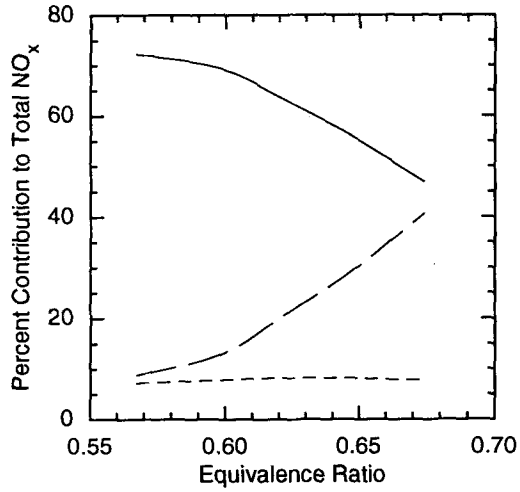

FIG. 3. PSR/PFR model prediction of the relative contribution of the thermal (— —), nitrous oxide (—) and prompt (---) NO_x pathways to the total amount of NO_x for the 10 atm, 615 K inlet temperature methane/air flame shown in Fig. 1. The sum of the three pathways is slightly less than 100% because other NO_x formation pathways contribute.

Conclusions

Chemical mechanisms of NO_x and CO formation and destruction have been analyzed under lean, premixed conditions representative of conditions in a gas turbine. New rate constants, reactions, and species were identified.

The Zeldovich mechanism apparently proceeds via the electronically excited $O(^1D)$ state, not by direct reaction with ground-state $O(^3P)$. Given the abundance of N_2 during combustion with air, reactions which produce $O(^1D)$ will almost certainly lead to NO formation. Inclusion of $O(^1D)$ chemistry in addition to $O(^3P)$ reactions may be necessary for accurate NO_x predictions.

Premixed gas turbine combustors attempting to achieve low NO_x emissions will operate under fuel-lean conditions near the blowout limit. Under these conditions, the nitrous oxide pathway will be the dominant route to NO_x formation, and almost all of the NO_x will form in the flame zone. Thermal NO_x becomes increasingly important at higher temperatures. Prompt NO_x is relatively unimportant for methane and ethane under lean conditions, but it becomes the dominant route to NO_x under fuel-rich

conditions. Kinetics of the nitrous oxide pathway will ultimately be the limiting factor that determines the lowest achievable NO_x emissions from premixed combustors.

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COMMENTS

Dr. Keith Schofield, University of California, Santa Barbara, USA. Your suggestion is quite intriguing of an alternative endothermic chemical quenching channel for $O(^1D) + N_2$ leading to NO that turns on at higher temperatures. Although there is no information in the literature for this, there is a significant amount for the reverse reaction. This N + NO reaction has been extensively studied and is a classical kinetic tool either for titrating N atoms, or for producing known amounts of $O(^3P)$. As far as my memory prevails, no complications due to the presence of $O(^1D)$ have ever been suggested in the literature concerning this reaction and such considerations have been made. Consequently due to microscopic reversibility I feel your suggestion must remain highly speculative until supported by additional data. Kinetic models can have considerable flexiblility and alternative schemes could be considered to reproduce your data.

Joseph W Bozzelli, New Jersey Institute of Technology, USA. Is there evidence for the rapid conversion of ^1N_2O adduct formed from O^1D + NO to the ^3N_2O, without collision. This type of reaction,

singlet to triplet conversion, is usually several orders of magnitude slower[1] than the 10^{13} /sec rate which may be needed, if this reaction analysis is to be applied to experimental studies at low pressures.

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Author's Reply. $O(^1D) + N_2 \leftrightarrows N(^4S) + NO(^2\Pi)$ is proposed in this work based on theoretical estimates which assume fast intersystem crossing. Chemically activated $N_2O(^1\Sigma)^*$ will be formed without crossing, but fast crossing from it to an $N_2O(^3\Pi)^*$ state would be required to form $^4N + {}^2NO$. If the crossing is slow, the channel would be less important or unimportant. If the crossing is fast, at low temperatures the (a) NO channel would be minor relative to (b) stabilization and to (c) reactive quenching, forming $O(^3P) + N_2$. For example, at 1000K $k_a = 3 \cdot 10^7$, $k_b = 3.5 \cdot 10^{10}$, $k_c = 1.9 \cdot 10^{11}$, although at 2000K, $k_a = k_c = 7 \cdot 10^{10}$.

N + NO studies to date do not support high $O(^1D)$ production. Felder and Young[1] experimentally compared the two-step mechanism $N + NO \leftrightarrows O(^1D) + N_2 \leftrightarrows O(^3P) + N_2$ to the direct $O(^1D) + N_2 \leftrightarrows O(^3P) + N_2$ reaction by attempting to remove $O(^1D)$ chemically. They found an upper limit of 10% contribution by the $O(^1D)$ mechanism, contrasting with our estimate of $O(^1D)$ domination by four orders of magnitude. Walch and Jaffe[2] did not examine curve-crossing to ^1N_2O, but their theoretical study of the triplet potential energy surfaces showed only a small barrier for $^4N + {}^2NO$ to a ^3N_2O surface that would correlate to $O(^3P) + N_2$.

Singlet-triplet intercrossing is necessary for $^3O + N_2 (+M) \leftrightarrows {}^1N_2O (+M)$, although the involvement of a third body may not only be for energy transfer (the basis of classical falloff) but also for collisionally medicated curve-crossing.

Agreement of a prediction with accurate data is a necessary but not sufficient condition for the model's correctness. We look forward to further theoretical and experimental work that will establish whether NO can be formed from $O(^1D)$.

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J. M. Beér, Massachussetts Institute of Technology, USA. You showed good agreement of calcualated and experimental data. Previous studies using PSR-PFR model sequencing have shown the results to be sensitively dependant on the split of the residence time between the two reactor types. How did you decide on the respective fractions of the residence time in the stirred and plug flow reactor?

Author's Reply. The model-predicted NO_x is very sensitive to the fraction of the total residence time spent in the PSR. NO_x emissions tend to increase as the PSR residence time (or volume) increases. We chose the PSR volume to match the experimentally observed lean blowout limit for a given combination of total mass flow, pressure, and inlet temperature. This choice yields very small PSR volumes and residence times the PSR volume was less than 1% of the total volume in all cases.

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A. Fontijn, Rensselaer Polytechnic Institue, USA. Your suggestion of a role of $O(^1D)$ in NO formation is intriguing. If correct would the $O(^1D)$ then not play a role in other flame chemistry processes as well, and is that compatible with further observations?

Author's Reply. Any $O(^1D)$ will react rapidly in flames, either consuming its co-reactant or transferring energy to it by inelastic (reactive) or elastic (physical) quenching to $O(^3P)$. By comparing k_x'B, ^1O should be to ^3O approximately as $^1CH_2$ is to $^3CH_2$. To generalize, the singlet would be expected to have barrierless insertions into π and σ bonds, giving equally fast kinetics with molecules or radicals. The triplet biradical reacts with molecules by somewhat slower H-atom abstractions and additions to π bonds, but it may also react with other radicals by faster combinations.

However, thermal ratios of $[^1O]/[^3O]$ are much lower than those of $[^1CH_2]/[^3CH_2]$ because of the larger singlet-triplet energy difference, 45.3 vs. 9.1 kcal/mol. $^1CH_2$ reactions can be more important than $^3CH_2$ reactions at flame temperatures because of faster singlet kinetics and adequate singlet concentrations, only on order of magnitude less than $[^3CH_2]$. For O-atom, thermal amounts of ^1O are still five orders of magnitude lower than $[^3O]$ at 2000K.

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Philip C Malte, University of Washington, USA. Have pressures other than 5 and 10 atm been studied with the mechanism? NO_x formation via nitrous oxide is sensitive to pressure. Thus, for aeroderivative gas turbine engines with lean-premixed com-

bustors, of pressures greater than 10 atm to about 30 atm, the formation of NO_x via nitrous oxide should be even more significant than shown for the 5 to 10 atm range. Has the NO_x sensitivity to pressure been examined? On the other hand, for 1 atm lean-premixed combustion, the nitrous oxide pathway may not be so significant. For every lean combustion at 1 atm, with temperatures in the 1500 to 1600K range, the prompt NO_x may be predominant. This appears to be the case the jet-stirred reactors.[1,2]

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Author's Reply. We have used our proposed mechanism to calculate premixed laminar flames structures at pressures from 1 to 40 atm. PSR compositions have also been calculated over this pressure range.

We also find that given the appropriate conditions (P = 1 atm, very fuel-lean & low-temperature, short residence time), the prompt NO_x pathway can be the primary route to NO_x in a PSR. Under these conditions, both the original Miller and Bowman mechanism and the modified mechanism proposed here predict that the prompt NO_x pathway can contribute 20–80% of the total NO_x. This observation highlights the difference between lean premixed atmospheric-pressure combustion and gas turbine combustion, where the N_2O pathway predominates.

In stirred reactors at a very high pressure (P = 40 atm), we again find that NO_x formation is dominated by the N_2O pathway, However, as pressure increases from 1 to 40 atm, the relative contribution of the O-atom reaction:

$$N_2 + O\,(+M) = N_2O\,(+M)$$

to N_2O formation decreases, while the relative contribution of the HO_2 reaction:

$$N_2 + HO_2 = N_2O + OH$$

increases. Under appropriate conditions Reaction (2) is predicted to account for ~40% of N_2O formation at 40 atm, versus ~2% at 1 atm. The increasing importance of Reaction (2) is due, in part, to an approximately order-of-magnitude increase in HO_2 concentration coupled with more than an order-of-magnitude decrease in O-atom concentration.

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Douglas A Greenhalgh, Cranfield Institute of Technology U.K. Generally the agreement for NO prediction against experiment by your model is very good except for one case, 300K inlet with CH_4/air. Why is this?

Author's Reply. The lower-pressure, non-preheated flames generally have the lowest mass flow rates and greatest relative heat loss, and they are furthest from adiabatic. We find that the PSR/PFR approximation works best when the system is closest to adiabatic, as is the case for the higher-pressure, higher mass flow, preheated flames. The reason for the poorer agreement in the flames with the greatest heat loss in not completely clear, but it may reflect experimental uncertainty in the determination of the heat loss (i.e., the temperature measurements and the energy balance).